

Joseph Jimin Kim

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EDUCATION

University of Pennsylvania, GSE Philadelphia, PA	August 2026 – May 2028
Accelerated Masters in Learning Science & Technologies	
Haverford College Haverford, PA	August 2021 – May 2027
Bachelors in Computer Science	
GPA: 4.0	
Honors and Awards: John P. Chesick Scholar; The Thomas Wistar, Sr. 1898 Memorial Scholar	

PROJECTS

[T\(AI\)](#) | *Python, Typescript, fastAPI, Next.js, PostgreSQL, Cloudflare R2*

- An educational RAG application with document-based conversations, summarization, and practice exams
- Designed a scalable RAG pipeline with asynchronous ingestion and parent-child chunking
- Implemented efficient retrieval with HNSW-based index and response with chunk citations

[Language Learning Buddy](#) | *Typescript, Python, React, Next.js, GraphQL, MongoDB*

- A personalized AI language learning platform with conversation and speech practice
- Fine-tuned LLMs on 6,000+ synthetic conversations using LoRA adapters for 3 languages
- Optimized LLM deployment with quantization and deployed real time chat through GraphQL subscription

[Bible Verse Explorer](#) | *Python, Javascript, Flask, PostgreSQL, Three.js*

- An interactive 3D web application to explore the semantic relationships between bible verses

WORK EXPERIENCE

Information Security Specialist Oct. 2024 – July 2026

Intelligence Information Systems Management Group, Republic of Korea Air Force

- Monitored external and internal network activity in headquarters of Korean Air Force
- Provided timely solutions to security breaches and system vulnerabilities
- Developed automated tools for routine checks for firewall and antivirus software, reducing manual workload

Research Intern Mar. 2024 – Sept. 2024

AIEd Lab, Yonsei University

- Led the development of an AI program improve student attention for recorded lectures using openCV library
- Investigated scaffolding approaches to visual-auditory substitution learning

Teaching Assistant & Lab Monitor Aug. 2022 – Jan. 2024

CS Department, Haverford College & Bryn Mawr College

- Helped instruct Introduction to CS and Discrete Mathematics, holding weekly office hours and grading homework
- Resolved technological difficulties and assisted in debugging Python and Java code for 40+ students

Undergraduate Researcher May 2023 – Nov. 2023

Summer Science Research, Bryn Mawr College

- Conducted research on the impact of various deadline policies in college level CS courses
- Reviewed relevant literature, designed survey, and analyzed data from 50+ participants
- First author of a peer-reviewed paper presented at the 2024 ACM SIGCSE Technical Symposium

NLP Research Assistant Jan. 2023 – Aug. 2023

Grissom Lab, Haverford College

- Collected and cleaned Korean-English corpus of more than 1M+ lines for NLP research
- Implemented pipelines to train Transformer-based MT models using Sockeye framework
- Evaluated the accuracy of various models and generated plots to represent key findings